Algorithm 2: Differentially Activated Feature Extraction: Cumulative Activations

Input: LLM $\mathcal{F}$, SAE Encoder $\mathcal{S}_{l,enc}$, layer index l, datasets $\mathcal{T}, \mathcal{T}^*$, integer K
Output: Set of differentially activated SAE feature indices
Initialize:
  $\mathbf{S}, \mathbf{S}^* \leftarrow \{\}, \{\}$;
  $\mathbf{z}_{cum}, \mathbf{z}^*_{cum} \leftarrow 0, 0$
for $i \leftarrow 0$ **to** $K-1$ **do**
  $\mathbf{H}_l, \mathbf{H}^*_l \leftarrow \mathcal{A}_l(\mathcal{F}(\mathcal{T}_i)), \mathcal{A}_l(\mathcal{F}(\mathcal{T}^*_i))$
  $\mathbf{Z}_l, \mathbf{Z}^*_l \leftarrow \mathcal{S}_{l,enc}(\mathbf{H}_l), \mathcal{S}_{l,enc}(\mathbf{H}^*_l)$
  $\mathbf{z}_{l,\text{last}}, \mathbf{z}^*_{l,\text{last}} \leftarrow \mathbf{Z}_l[-1], \leftarrow \mathbf{Z}^*_l[-1]$;
  $\mathbf{z}_{cum} \leftarrow \mathbf{z}_{cum} + \mathbf{z}_{l,\text{last}}$;
  $\mathbf{z}^*_{cum} \leftarrow \mathbf{z}^*_{cum} + \mathbf{z}^*_{l,\text{last}}$;
indices, indices* $\leftarrow$ TopK$(\mathbf{z}_{cum}, K)$, TopK$(\mathbf{z}^*_{cum}, K)$;
return indices* $\setminus$ indices

ones. It begins by initializing two zero vectors, $\mathbf{z}_{cum}$ and $\mathbf{z}^*_{cum}$, to serve as accumulators. Within the loop over the datasets, after the final-token latent vectors ($\mathbf{z}_{l,\text{last}}$ and $\mathbf{z}^*_{l,\text{last}}$) are extracted for each instance pair, they are summed into their respective cumulative vectors. This process continues for all K instances. Only after the loop is complete are the top-K feature indices extracted from the final summed vectors $\mathbf{z}_{cum}$ and $\mathbf{z}^*_{cum}$. This method hypotheses that the importance of a feature is captured not only by how often it appears but also by its average activation magnitude across the entire dataset. The final set of discriminative features is again found by taking the set difference between the indices from the augmented and standard datasets.

D SAE Feature Distribution

We report the SAE feature distribution, and then show the tables of the top activated features for the chain-of-thought-math reasoning cases, as well as for some instruction following cases. The Figures are shown from Figure 6 to Figure 17.

E SAE Feature and Their Explanations

E.1 Chain-of-thought math reasoning

We typically set $K = 10$. Results for the Gemma-2 model family, including Gemma-2-2B, Gemma-2-9B, and Gemma-2-9B-it, are presented in the main findings. The corresponding sparse autoencoders are pre-trained in Gemma-Scope. The tables below show the top 5 most activated SAE features (out of the top 10). We define F^k_{cot} as the set of k most active features on Chain-of-Thought (CoT) inputs, and F^k_{std} as the set of k most active features on standard inputs without CoT.

Further, we define the *SAE-cardinality*(SC) as the cardinality of the difference between the set F^k_{cot} and the set F^k_{std}:

$$\text{SC} = \left| F^k_{cot} \setminus F^k_{std} \right| \tag{6}$$

For Gemma-2-2b,

$$F^5_{std} = \{6868, 12748, 2221, 10640, 9761\}$$

$$F^5_{cot} = \{6631, 15153, 1858, 1642, 2899\}$$

$\Rightarrow$ Gemma2-2b-SC$_5$ $= \left|F^5_{cot}\right| = 5$.

For Gemma-2-9b,

$$F^5_{std} = \{1126, 1744, 1877, 1935, 11115\}$$

$$F^5_{cot} = \{7007, 1126, 9005, 6782, 12946\}$$

$\Rightarrow$ Gemma2-2b-SC$_5$ $= |\{7007, 9005, 6782, 12946\}| = 4$.

For Gemma-2-9-it,

$$F^5_{std} = \{5679, 1426, 2492, 2995, 10044\}$$

$$F^5_{cot} = \{2995, 6096, 13390, 15219, 2967\}$$

$\Rightarrow$ Gemma2-2b-SC$_5$ $= |\{6096, 13390, 15219, 2967\}| = 4$.

We use this metric to avoid steering redundant or meaningless SAE features. For example, in Gemma-2-9B models, we remove feature 1126, described as *data in structured formats*. The model instead identifies its two most active features as **terms related to automotive diagnostics and their connection methods** and **phrases indicating reasons**, which align with Chain-of-Thought prompting. (In further experiments, we show that the top activated SAE feature 7007 does not perform as well as SAE feature 6782.) For the instruction-tuned Gemma-2-9B model, feature 2995, described as *different types of data structures*, is removed. This is similar to the top active but unhelpful feature 1126 in the base Gemma-2-9B model. As a result, the model now surfaces top active features *interpreted as* **phrases indicate differing interpretations** and **phrases related to legal analysis and reasoning**, both of which suggest the model is engaging in Chain-of-Thought reasoning.

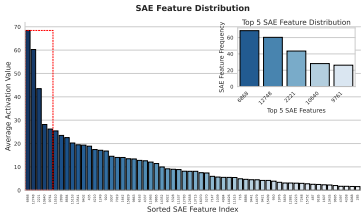

Figure 6: Gemma-2-2b (w/o CoT)

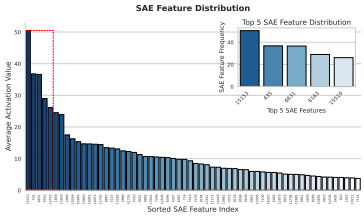

Figure 7: Gemma-2-2b (w/ CoT)

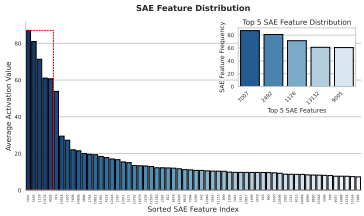

Figure 8: Gemma-2-9b (w/o CoT)

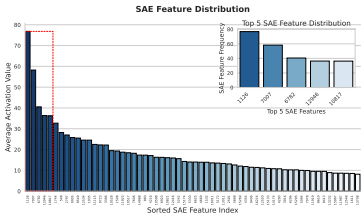

Figure 9: Gemma-2-9b (w/ CoT)

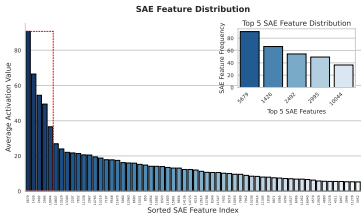

Figure 10: Gemma-2-9-it (w/o CoT)

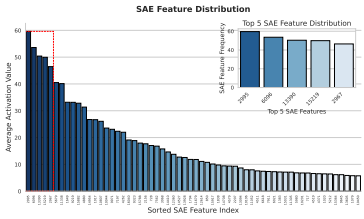

Figure 11: Gemma-2-9-it (w/ CoT)

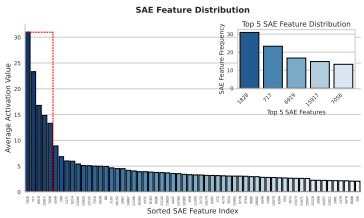

Figure 12: Gemma-2-9b-it (w/o instr.) on JSON

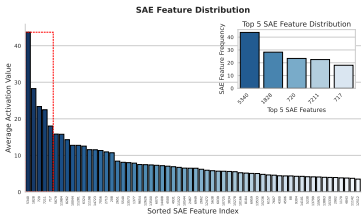

Figure 13: Gemma-2-9b-it (w/ instr.) on JSON

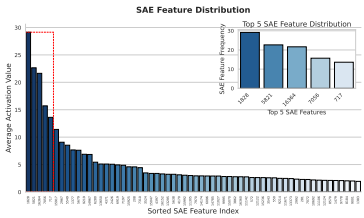

Figure 14: Gemma-2-9b-it (w/o instr.) on Lowercase

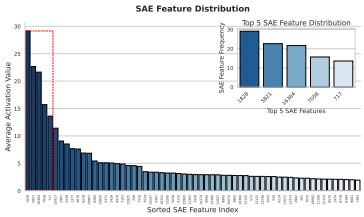

Figure 15: Gemma-2-9b-it (w/ instr.) on Lowercase

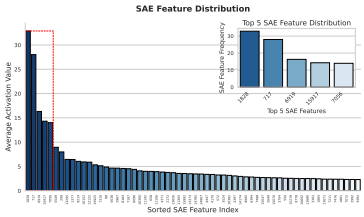

Figure 16: Gemma-2-9b-it (w/o instr.) on Uppercase

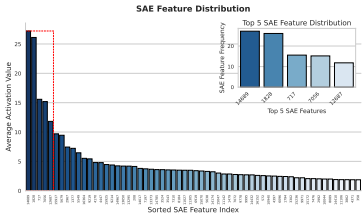

Figure 17: Gemma-2-9b-it (w/ instr.) on Uppercase

Rank	Index	Feature Description	Activation
1	6631	the beginning of a text or important markers in a document	63.25
2	15153	mathematical expressions and relationships	35.24
3	1858	punctuation marks and sentence endings	33.51
4	1642	numeric values and their relationships in a technical context	27.37
5	2899	specific instructions or steps related to a process	24.51

Table 5: Top 5 activated SAE features for Gemma-2-2B with Chain-of-Thought

Generally we define F^k_{inst} as the set of k features active the most on Instruction Followed inputs, and F^k_{std} as the set of k features active the most on standard inputs without instructions. The steering experiments are performed under zero-shot setting, except for the length constraint case where instructions are appended to the input.

E.2 JSON Format

Feature-Level Activation Analysis To understand the effect of Instruction following behavior in JSON generation, we analyze the top activated SAE features between the Instruction following setting and the baseline (no instruction). Following the cumulative extraction method, we average

Rank	Index	Feature Description	Activation
1	1126	numerical data and references in structured formats	73.33
2	1744	mathematical terms and expressions	35.54
3	1877	technical or programming-related terms and functions	35.48
4	1935	sentences with numerical data and measurements related to various subjects	30.99
5	11115	dates and structured data representations	24.01

Table 6: Top 5 activated SAE features for Gemma-2-9B without Chain-of-Thought

Rank	Index	Feature Description	Activation
1	7007	terms related to automotive diagnostics and their connection methods	98.91
2	1126	numerical data and references in structured formats	88.26
3	9005	terms and phrases related to healthcare interventions and studies	71.25
4	6782	technical terms and processes related to scientific algorithms and methodologies	47.57
5	12946	phrases indicating reasons or justifications in legal or analytical contexts	28.53

Table 7: Top 5 activated SAE features for Gemma-2-9B with Chain-of-Thought

Rank	Index	Feature Description	Activation
1	5679	references to specific data science tools and technologies	90.71
2	1426	phrases with statistical or probabilistic references	66.50
3	2492	phrases related to processes and outcomes in negotiations or transactions	54.44
4	2995	different types of data structures and their attributes	49.43
5	10044	references to statistical analysis and methods in research contexts	36.55

Table 8: Top 5 activated SAE features for Gemma-2-2B-it without Chain-of-Thought

Rank	Index	Feature Description	Activation
1	2995	different types of data structures and their attributes	59.61
2	6096	phrases and contexts that indicate differing interpretations or meanings	53.60
3	13390	phrases and terms related to legal analysis and reasoning in court cases	50.40
4	15219	elements of mathematical expressions and functions	49.95
5	2967	technical terms and coding references related to programming and web development	46.46

Table 9: Top 5 activated SAE features for Gemma-2-9B-it without Chain-of-Thought

Rank	Index	Feature Description	Activation
1	1828	References to API requests and their parameters in code snippets	30.98
2	717	Structured elements within technical documentation	23.32
3	6919	Code snippets related to API data retrieval and error handling	16.78
4	15917	Formatted questions and responses	14.83
5	7056	Scientific concepts and terminologies related to health and wellness	13.31

Table 10: Top 5 activated SAE features for Gemma-2-9B-it without instruction (JSON)